BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.



RHS COMPUTING / OCR H446 / 1.4.3 LESSON 030 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 030 · Half adders and full adders

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When we group four neighbouring 1s on a Karnaugh map, what happens to the Boolean expression?

Q2. Represent −13 using 8-bit sign and magnitude.

Q3. Insert 40, 20, 60, 10, 30, 50 into an empty binary search tree. Draw the resulting tree.

Q4. Using a 5-bit two’s complement mantissa and 4-bit two’s complement exponent, calculate A − B. Show exponent alignment and mantissa subtraction. A = $0.1100_2 \times 2^3$; B = $0.1000_2 \times 2^2$.

# I Do · Read new knowledge

## Key vocabulary

Recall the term — half adder

Recall the term — full adder

Recall the term — Sum

Recall the term — Carry

Recall the term — carry-in

Recall the term — carry-out

Recall the term — XOR

## New learning

Start with binary addition — One binary column can produce a sum bit and a carry bit.

Meet the half adder — A half adder adds two single-bit inputs: A and B.

Discover the two gates — XOR calculates Sum; AND calculates Carry.

Build its truth table — Test every possible pair of inputs.

Find the limitation — What happens when a previous column sends a carry into this column?

First half adder — Add A and B to produce an intermediate Sum $S_1$ and Carry $C_1$.

Second half adder — Feed $S_1$ and Carry-in into a second half adder. Its Sum is the final Sum.

Combine both carries — Feed $C_1$ and $C_2$ into an OR gate to obtain Carry-out.

## Worked example

Model 1 + 1 with a half adder

1. Rightmost result bit — CODE:
Sum = 0

2. Next column — CODE:
Carry = 1

3. Trace XOR and AND — CODE:
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry

4. Connect both half adders — CODE:
$S_1$ = A XOR B







0 0 1 | 1 0
0 1 0 | 1 0
0 1 1 | 0 1
1 0 0 | 1 0
1 0 1 | 0 1
1 1 0 | 0 1
1 1 1 | 1 1
Check: The outputs match ordinary binary addition of three bits.

Q3. Explain four-bit addition
CODE:
Use one full adder for each bit position. The carry-out from a lower-order full adder is connected to the carry-in of the next higher-order full adder. This propagates carries across the four-bit addition.
Check: Your answer connects carry-out from one bit to carry-in of the next.

Q4. Add 0111 + 0001
CODE:
0111
+ 0001
------
1000

A carry is generated at the least-significant column and propagates through the next two columns.
Check: The final result is 1000.

## Exit Ticket · Model answer

True. A half adder uses XOR for Sum and AND for Carry.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. The Boolean expression becomes simpler. Variables that change within the group can be removed, leaving only variables that stay the same.

Q2. Step 1: 13 in binary is 1101. Step 2: pad the magnitude to seven bits: 0001101. Step 3: because the number is negative, put 1 in the sign bit. Answer: 1 | 0001101 = 10001101. This is sign and magnitude , not two’s complement. SIGN · 1 BIT 1 Negative MAGNITUDE · 7 BITS 0001101 13 in binary

Q3. Insert the values in the order given . 40 becomes the root. 20 goes left of 40; 60 goes right of 40. 10 goes left of 20; 30 goes right of 20; 50 goes left of 60. Every smaller value goes to the left and every larger value to the right. 40 20 60 10 30 50

Q4. Align exponents: B = $0.0100_2 \times 2^3$. Subtract mantissas: $0.1100_2 - 0.0100_2 = 0.1000_2$. Answer: $0.1000_2 \times 2^3 = 4$.

### You Do pages · Independent questions

Q1. Half-adder question
CODE:
S = A XOR B
C = A AND B

A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1

Two half adders can be joined, with an OR gate combining carry outputs, to form a full adder.
Check: Use XOR for Sum, AND for Carry and mention the OR connection.

Q2. Complete eight cases
CODE:
A B Cin | S Cout
0 0 0 | 0 0





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Sum = $S_1$ XOR Cin
Cout = (A AND B) OR ($S_1$ AND Cin)

## We Do · Practise with guidance

Half-adder expressions

OCR asks for S and C.

Complete the same formulae

```
S = A ? B
C = A ? B
```

```
S = A XOR B
C = A AND B
```

Half-adder truth table

Complete the four outputs.

Fill the same table

```
A B | S C
0 0 | ? ?
0 1 | ? ?
1 0 | ? ?
1 1 | ? ?
```

```
A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1
```

From half to full adder

Trace the two XOR gates, then the two AND gates feeding OR.

Improve the same explanation

```
Two half adders plus an ? gate can form a full adder.
```

```
Two half adders plus an OR gate can form a full adder.
```

Full-adder case

A=1, B=1, CarryIn=1.

Complete the output

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Half-adder question
State the Sum/Carry expressions, complete the truth table, then explain how half adders contribute to a full adder.

### Question 2

Complete eight cases
Without looking at the diagram, sketch the five gates and their connections from memory. Then create the full-adder truth table for A, B and CarryIn.

### Question 3

Explain four-bit addition
Explain how four full adders can add two 4-bit binary numbers.

### Question 4

Add 0111 + 0001
Use bit-by-bit addition and identify where the carry travels.

## Exit Ticket · Answer before checking



RHS COMPUTING / OCR H446 / 1.4.3 LESSON 030 · DRAFT
BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

For a half adder, Sum is produced by XOR and Carry is produced by AND.